

Answers & Mark Scheme — Future Stories (FSCE)-Style Sample Paper — Adventure & Beacon Maths

Total marks: 40

Section 1: Multiple Choice

Q1 A. 47 [1]

XL = 40 and VII = 7, so XLVII = 40 + 7 = 47.

Q2 A. Lakeside [1]

The coldest temperature is the lowest number: -9°C at Lakeside is lower than -6°C , -2°C and 3°C .

Q3 A. 68000 [1]

The digit in the hundreds place is 4, which is less than 5, so 68,472 rounds down to 68,000.

Q4 A. 500 [1]

The digit 5 is in the hundreds column of 803,516, so its value is $5 \times 100 = 500$.

Q5 A. 7°C [1]

$-4 + 11 = 7$, so the midday temperature was 7°C .

Q6 A. 1948 [1]

M = 1000, CM = 900, XL = 40, VIII = 8, so MCMXLVIII = 1000 + 900 + 40 + 8 = 1948.

Q7 A. 38 [1]

Brackets first: $9 - 4 = 5$. Then multiplication: $6 \times 5 = 30$. Then addition: $8 + 30 = 38$.

Q8 A. 14 [1]

$84 \div 14 = 6$ exactly, so 14 is a factor of 84. 84 does not divide exactly by 9, 16 or 25.

Q9 A. 61 [1]

$51 = 3 \times 17$, $57 = 3 \times 19$ and $63 = 7 \times 9$, so all three have factors other than 1 and themselves. 61 has no factors other than 1 and 61, so it is prime.

Q10 A. 49 [1]

The n th square number is $n \times n$. The 7th square number is $7 \times 7 = 49$.

Q11 A. 125 [1]

$5^3 = 5 \times 5 \times 5 = 125$.

Q12 A. 9/12 [1]

$3/4 = 9/12$ because $3 \times 3 = 9$ and $4 \times 3 = 12$, so both the numerator and denominator have been multiplied by the same amount.

Q13 A. 0.35 [1]

$7/20 = 35/100 = 0.35$.

Q14 A. 0.3 [1]

As decimals: $3/8 = 0.375$, $2/5 = 0.4$, $1/3 = 0.333\dots$, and 0.3 is already a decimal. 0.3 is the smallest of the four.

Q15 A. 5/12 [1]

The lowest common denominator of 4 and 6 is 12: $\frac{1}{4} = \frac{3}{12}$ and $\frac{1}{6} = \frac{2}{12}$, so $\frac{3}{12} + \frac{2}{12} = \frac{5}{12}$, which is already in its simplest form.

Q16 A. 4.3 [1]

The digit in the hundredths place is 6, which is 5 or more, so the tenths digit rounds up from 2 to 3, giving 4.3.

Q17 A. 3 : 4 [1]

The highest common factor of 18 and 24 is 6. $18 \div 6 = 3$ and $24 \div 6 = 4$, so the simplest form is 3 : 4.

Q18 A. 500 g [1]

$200 \div 4 = 50$ g per person. $50 \times 10 = 500$ g for 10 people.

Q19 A. 6 [1]

In the altos : basses ratio (4 : 3), 8 altos corresponds to 4 parts, so 1 part = 2. Basses = 3 parts = $3 \times 2 = 6$.

Q20 A. 57° [1]

Angles on a straight line add up to 180°. $180 - 55 - 68 = 57^\circ$.

Q21 A. 56° [1]

Angles in a triangle add up to 180°. $180 - 90 - 34 = 56^\circ$.

Q22 A. It has 5 faces [1]

A cube has 6 faces, 8 vertices and 12 edges, so 'It has 5 faces' is the one statement that is not true.

Q23 A. 6 [1]

A regular hexagon has one line of symmetry through each pair of opposite vertices (3 lines) and one through each pair of opposite edges (3 lines), giving 6 in total.

Q24 A. (5, 3) [1]

Moving 3 right adds 3 to the x-coordinate: $2 + 3 = 5$. Moving 2 down subtracts 2 from the y-coordinate: $5 - 2 = 3$. So Q is at (5, 3).

Q25 A. 13 [1]

Thursday = 31 books, Monday = 18 books. $31 - 18 = 13$ more books.

Q26 A. 15 [1]

Walk = 34 pupils, bus = 19 pupils. $34 - 19 = 15$ more pupils walk than travel by bus.

Q27 A. 10am and 12pm [1]

The readings rise by 3°C (6-8am), 2°C (8-10am), 6°C (10am-12pm), then 1°C (12-2pm) before falling. The biggest rise, 6°C, is between 10am and 12pm.

Q28 A. 10 [1]

$9 + 6 + 5 = 20$ pupils have a cat and/or a dog. $30 - 20 = 10$ pupils have neither.

Section 2: Short Written Answers

Q1 8024 [1]

$236 \times 34 = 236 \times 30 + 236 \times 4 = 7080 + 944 = 8024$.

Q2 16 [1]

$187 \div 12 = 15$ remainder 7, so 15 minibuses only seat 180 pupils; a 16th minibus is needed for the remaining 7 pupils.

Q3 22 kg [1]

$128 + 96 + 154 = 378$ kg collected in total. $400 - 378 = 22$ kg short of the target.

Q4 6 km [1]

$15 \div 5 = 3$, then $3 \times 2 = 6$ km.

Q5 800 [1]

$25\% = 1/4$, and $3,200 \div 4 = 800$.

Q6 £102 [1]

10% of £120 = £12, and 5% = £6, so 15% = £18. $£120 - £18 = £102$.

Q7 £200 [1]

$4 + 5 = 9$ parts in total. $£360 \div 9 = £40$ per part. Priya's share = $5 \times £40 = £200$.

Q8 32 km [1]

$6.4 \times 5 = 32$ km.

Q9 40 m [1]

Going around the shape: $12 + 5 + 5 + 3 + 7 + 8 = 40$ m. The two unlabelled sides are found from $8 - 3 = 5$ m and $12 - 5 = 7$ m.

Q10 198 m² [1]

Area = length $\times$ width = $18 \times 11 = 198$ m².

Q11 10 minutes [1]

$8 + 11 + 6 + 13 + 12 = 50$ minutes in total. $50 \div 5 = 10$ minutes.

Q12 12:35 [1]

$09:45 + 2$ hours = 11:45. $11:45 + 50$ minutes = 12:35.